

Gun Violence Prediction in Philadelphia

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MUSA 6500 Deep Learning in Python

§1 Background

§2 Modeling Strategy

§3 Findings

Global Regression

GWPR

§4 Discussions,
Limitations, and
Conclusion

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§3 Findings

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§4 Discussions,
Limitations, and
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- ▶ Gun violence in Philadelphia has been a persistent and deeply complex crisis, affecting thousands of residents each year and leaving a profound impact on victims, families, and entire communities.
- ▶ Nearly 10,000 shootings since 2019, resulting in over 11,000 victims.
- ▶ The burden of gun violence in Philadelphia is not evenly distributed in space.

Takeaway: Timely and data-driven resource allocation is crucial for effective gun violence prevention in Philadelphia!

§1 Background

§2 Modeling Strategy

§3 Findings

Global Regression

GWPR

§4 Discussions,
Limitations, and
Conclusion

§2 Modeling Strategy

Model choice: Geographically-Weighted Poisson Regression

- ▶ To account for spatial patterns: Shootings exhibit strong spatial autocorrelation and clustering, rather than randomly distributed across the city.
- ▶ Spatial unit of analysis: Police districts
 - ▶ Commonly used for resource allocation, law enforcement deployment, and public safety interventions.
 - ▶ Findings are directly actionable for city agencies and stakeholders.
- ▶ Poisson regression model: For handling count data. (The number of shootings within each district and time period)

Overall, the model can account for local variations in the relationships between predictors and shooting counts, effectively capturing spatial heterogeneity and dependence.

To predict shooting counts, the following covariates are included:

- ▶ 1-lagged Kernel Density Estimate (KDE): Measures the intensity of shootings in the previous time period, capturing spatial and temporal autocorrelation.
- ▶ 1-lagged Normalized Difference Vegetation Index (NDVI): Proxy for green space and environmental quality, which have been linked to crime rates.
- ▶ 1-lagged Normalized Difference Built-up Index (NDBI): Reflects urbanization and built environment, another factor associated with crime patterns.
- ▶ Year of Shooting: Controls for long-term trends and policy changes over time.
- ▶ Quarter of Shooting (categorical): Captures seasonal effects and temporal fluctuations within each year.

To assess the contribution of environmental covariates and spatial predictors, two models are estimated:

- ▶ Model 1 (Baseline): This model uses only 1-lagged KDE, year of shooting, and quarter of shooting as covariates. This model evaluates the explanatory power of recent shooting intensity and temporal controls, serving as a benchmark for spatial autocorrelation effects.
- ▶ Model 2 (Extended): This model adds the 1-lagged NDVI and NDBI to the predictors in Model 1. This model tests whether incorporating environmental and urbanization measures improves predictive accuracy and provides additional insights into the determinants of gun violence.

This modeling approach is grounded in [crime pattern theory](#) and [environmental criminology](#), which emphasize the importance of the following factors in shaping crime risk:

- ▶ Spatial context
- ▶ Routine activities
- ▶ Environmental features

By integrating spatially explicit predictors and leveraging advanced statistical techniques, the analysis aims to uncover actionable patterns that can inform targeted interventions and resource allocation.

§1 Background

§2 Modeling Strategy

§3 Findings

Global Regression

GWPR

§4 Discussions,
Limitations, and
Conclusion

§3 Findings

Variable	Estimate	<i>p</i> -value
Intercept	71.807	0.000
1-lagged KDE	0.408	0.000
Year	−0.034	0.000
2nd Quarter	0.124	0.000
3rd Quarter	0.159	0.000
4th Quarter	0.049	0.140

Table 1: Model 1 Results. Adjusted percent deviance explained: 37.8%.

Variable	Estimate	<i>p</i> -value
Intercept	60.585	0.000
1-lagged KDE	0.459	0.000
1-lagged NDVI	0.365	0.000
1-lagged NDBI	0.245	0.000
Year	−0.029	0.000
2nd Quarter	0.111	0.004
3rd Quarter	0.239	0.000
4th Quarter	0.105	0.005

Table 2: Model 2 Results. Adjusted percent deviance explained: 42.7%.



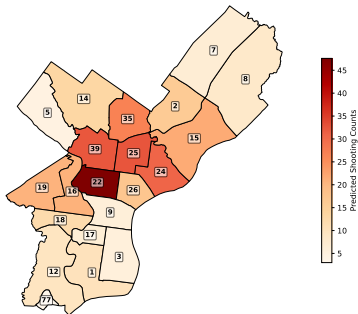
Variable	Mean	SD	Minimum	Median	Maximum
Intercept	63.896	65.496	-44.528	50.469	184.341
1-lagged KDE	0.678	0.358	0.155	0.731	1.250
Year	-0.030	0.032	-0.090	-0.023	0.023
2nd Quarter	0.136	0.083	-0.010	0.138	0.249
3rd Quarter	0.155	0.090	0.002	0.142	0.327
4th Quarter	0.056	0.101	-0.179	0.090	0.236

Table 3: Summary Statistics for GWPR Parameter Estimates for Model 1.

Variable	Mean	SD	Minimum	Median	Maximum
Intercept	65.503	92.946	-116.256	75.854	247.235
1-lagged KDE	0.634	0.452	0.061	0.493	1.598
1-lagged NDVI	0.290	0.627	-1.457	0.234	1.303
1-lagged NDBI	0.109	0.170	-0.257	0.135	0.440
Year	-0.031	0.046	-0.121	-0.036	0.059
2nd Quarter	0.201	0.357	-0.687	0.123	0.878
3rd Quarter	0.158	0.214	-0.280	0.157	0.491
4th Quarter	0.007	0.239	-0.484	0.059	0.430

Table 4: Summary Statistics for GWR Parameter Estimates for Model 2.

Predicted Shooting Counts by Police District (2025 Q1)



Predicted Shooting Counts by Police District (2025 Q1)

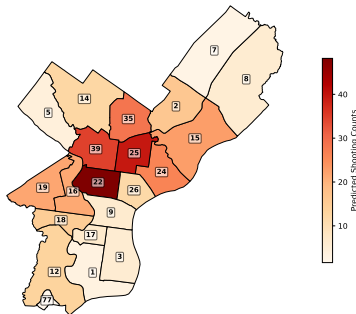


Figure 1: Predicted Maps of Shootings by Police Districts in 2025 Q1. Left: Model 1. Right: Model 2.

§1 Background

§2 Modeling Strategy

§3 Findings

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**§4 Discussions,
Limitations, and
Conclusion**

§4 Discussions, Limitations, and Conclusion

- ▶ **Performance Gains:** GWPR models outperform global regressions, indicating strong spatial heterogeneity in gun violence.
- ▶ **Baseline Model (Model 1):** 68.4% deviance explained (adjusted). Historical patterns and trends are strong predictors.
- ▶ **Extended Model (Model 2):** 78.0% deviance explained (adjusted). Environmental features (NDVI/NDBI) add predictive power.
- ▶ **Significance:** Nearly all covariates significant at 5% and 0.5% thresholds.
- ▶ **Hotspots:** North/Northeast Philadelphia (Districts 22, 25, 24, 39, 35, 15) identified as priority areas.

- ▶ **Event Rarity:** Shootings are rare, limiting granularity. Monthly data risks sparsity; quarterly data balances detail and reliability.
- ▶ **Covariate Scope:** Small sample size restricts model complexity. Results are preliminary, supporting but not proving environmental criminology.

In summary, while the models offer valuable insights into the spatial and environmental determinants of gun violence in Philadelphia, future research with richer datasets and additional covariates could further refine these findings and enhance their policy relevance.

Thank You!